MILITARY SPECIFICATION

FILM, INHIBITING, ETHYL CELLULOSE

This specification is approved for use by the Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inhibiting film manufactured for use as the peripheral inhibiting wrap for rocket propellant grains (Ballistite type).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-63462 - Plastic Molding and Extrusion Material, Ethyl Cellulose, for Rocket Grain Inhibiting Materials

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellant, Solid, Sampling, Examination and Testing
MIL-STD-1168 - Lot Numbering of Ammunition

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-LCA-OK, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 9330

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DRAWINGS

U.S. Army Missile Command

9240825 - Tape, Inhibiting (Ethyl Cellulose)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 569 - Measuring the Flow Properties of Thermoplastic Molding Materials
D 570 - Water Absorption of Plastics
D 648 - Deflection Temperatures of Plastics under Flexural Load
D 792 - Specific Gravity and Density of Plastics by Displacement
D 882 - Tensile Properties of Thin Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The material shall consist of ethyl cellulose and plasticizer(s) meeting the requirements of Type 1, MIL-P-63462. Suitable dyes or other colorants may be added where required to obtain colors specified by the procuring agency (see 3.6.2).

3.1.1 Plasticizer. The contractor shall supply the chemical name of each plasticizer used in the film, and samples shall be submitted to the procuring agency when requested.

3.1.2 Cellulose modification. The manufacturer shall provide the average percent of ethylation and the average viscosity (of a 5 percent solution of the flake in 60:40 toluene - Formula 2B Alcohol solvent mixture) of the ethyl cellulose used in making the determinations. Both determinations shall be run in triplicate on each lot of flake and the ranges reported.

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3.2 Method of manufacture. The film shall be made by the continuous extrusion method, or equivalent, as approved by the procuring agency.

3.3 Rework. Rework or scrap may be used if the resulting reworked film meets the test requirements listed herein. Film manufactured from scrap or rework shall be from the same source and of the same basic formulation as the virgin materials, but need not be from the same factory run or batch.

3.4 Dimensions. All dimensions and limits shall be as selected and requested by the procuring activity from the schedule of sizes shown on Drawing No. 9240825. All dimensional requirements are based on film conditioned and measured at 25 $\pm$ 1°C (77 $\pm$ 2°F) and 45 $\pm$ 5% relative humidity.

3.5 Roll characteristics.

3.5.1 Core. The film shall be tightly wound on a multilayer paper core with dimensions as shown on Drawing No. 9240825. The core shall not be collapsed during or after the winding operation.

3.5.2 Tightness of roll. Film rolls shall be tightly wound to prevent telescoping of the roll during handling and shipping operations (see 4.4.1.3.1).

3.5.3 Splices. Two splices shall be permitted on the 1372 meter (4500 foot) length rolls. No splices are permitted on the 457.2 meter (1500 foot) rolls (see 4.4.1.3.4).

3.5.4 Telescoped or crimped edges. The rolls of film shall be smoothly wound so that no film edges project or become crimped during winding or subsequent handling and packaging (see 4.4.1.3.6).

3.5.5 Time limits and storage conditions. Because of the nature of this material and the tendency for dimensions to change upon storage, the manufacturer shall only be held accountable for dimensions of his product for a period of 120 days after final slitting and winding of film rolls. Any other properties directly affected by dimensional changes are also subject to this time limitation. In order to minimize dimensional changes of the sample chosen for testing, the samples shall be stored and shipped under conditions within the following limits: Temperature, 10 to 37.8°C (50 to 100°F); Relative Humidity, 30 to 80%. The manufacturer will not be held accountable for the dimensions of his product under the 120 days limit imposed above after receipt of the product by the procuring agency unless the procuring agency stores the product under the temperature and humidity conditions listed above.

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3.6 Visual appearance.

3.6.1 Extrusion defects.

3.6.1.1 Fisheyes and gels. The film shall be relatively free from fisheyes or gels when inspected as indicated in 4.4.1.4.1.1.

3.6.1.2 Bubbles. No bubble or void in the body of the plastic into which a 1.59 millimeter (1/16 inch) diameter circle can be fitted is permissible. No elongated bubble or void with a length of 25.4 millimeters (1 inch) or more and a width of 0.25 millimeter (0.010 inch) or more is permissible. No group or cluster of pinpoint bubbles or bubble streaks with a cluster grouping into which a circle of 6.35 millimeters (1/4 inch) diameter can be fitted is permissible (see 4.4.1.4.1.2).

3.6.1.3 Striations. No striations are permissible that permit the film thickness to vary more than 0.013 millimeters (0.0005 inch) from the thickness of the film surrounding the striation, as determined by gage measurements at the point of striation and on the surrounding film (see 4.4.1.4.1.3).

3.6.1.4 Laminations. The film shall not contain laminations that can be detected by the inspection method described in 4.4.1.4.1.4.

3.6.2 Color. Unless otherwise specified by the procuring agency, the film shall be furnished uncolored. (See 3.6.2.1). When colored films are specified, the color shall match the sample furnished by the procuring agency.

3.6.2.1 Uncolored film. The natural color of the film shall be a light clear yellow or amber. A dark yellow or brown color is indicative of excessive degradation of the plastic resin during processing. The color of a single layer of film viewed against a white background under a daylight lighting shall be no darker than the color supplied by a combination of Kodak Color Compensating Filters CC05Y and CC10Y when viewed under the same conditions as the film. (See 4.4.1.4.2).

3.6.3 Clarity. The film shall have sufficient clarity to permit the nspection of inhibited grains for inhibiting defects (see 4.4.1.4.3).

3.7 Physical properties. The film shall conform to the requirements listed in Table I.

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TABLE I. Physical requirements.

Property	Requirement
Tensile strength, minimum, MPa (psi)	34 (4,950)
Elongation, minimum, %	28
Specific gravity @ 23°C/23°C	1.06 to 1.17
Water absorption, maximum, %	3.0
Soluble matter loss, maximum, %	0.2
Weight loss on heating, mg/cm^2	0.6
Deflection temperature, minimum, °C (°F)	54 (130)
Flow Temperature, °C (°F)	145-155 (293-311)

3.8 Migration characteristics. When four samples of film taken from separate film rolls are each tested in duplicate by the procedure described in 4.4.1.7, the migration properties of the film shall fall within the following limits. Duplicate tests are to be averaged for each roll and reported as a single value.

Average weight gain of 4 samples	20 to 24%
Range of weight gain (difference between highest and lowest of the 4 average sample results), maximum	2.5%
Average nitroglycerine gain of 4 samples	14 to 18%
Range of nitroglycerine gain (difference between highest and lowest of the 4 sample results, maximum	4%
Average inert gain of 4 samples (weight gain minus nitroglycerine gain)	4 to 8%

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3.9 Wrappability. Inhibiting films shall wrap satisfactorily when used to inhibit propellant grain (Ballistite type) as described in 4.4.1.7. The presence of stretched or warped film sections within a roll is likely to cause the film to wrap unsatisfactorily.

3.10 Adhesion. Inhibiting film shall adhere to the propellant grain with a bond strength of 5.85 MPa (850 psi) or greater, when tested as specified in 4.4.1.8. This requirement is not applicable to Radford Army Ammunition Plant.

3.11 Modified Taliani stability. When tested according to the procedure in Section 4.4.1.9, samples of the ethyl cellulose film and M-5 propellant shall not exhibit a rise in pressure greater than 200 mm corrected with EC blank value, and the ratio of the system used should be corrected to the volume basis (12 ml).

4. QUALITY ASSURANCE PROVISIONS

4.1 Lots.

4.1.1 First article inspection. Inhibiting film shall be manufactured from the same lot or batch of plastic granules that was qualified for production. The production shall be on a continuous basis and testing shall be conducted until requirements of the specification are met.

4.1.2 Production lot. Unless otherwise specified, a production lot shall consist of inhibiting film produced in a single continuous run or extrusion of film which the film rolls are cut or slit according to Drawing No. 924085 and prepared as an individual shipment. Maximum size of a production lot shall be limited to 27,216 kilograms (60,000 pounds) of material. When material lots or major operating conditions are changed, a new production lot shall be designated in accordance with the practice outlined in MIL-STD-105.

4.2 Inspection unit. For purpose of acceptance or rejection in accordance with this specification and MIL-STD-105, an inspection unit shall be a finished roll of film.

4.3 Sampling.

4.3.1 Sampling first article and periodic production acceptance tests. Fifty rolls shall be submitted as a preproduction lot and fifty rolls shall be selected at random during production of each 50,000-roll increment of a manufacturer's contract production to be sent to a testing activity designated by the procuring agency for tests in accordance with Table II. Acceptance or rejection criteria of the lot tested shall be in accordance with the rejection requirements of this table. This

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paragraph is not applicable to production of inhibiting film at the Radford Army Ammunition Plant, for which the following will apply: When two rolls of inhibiting film meet the requirements of Table III, they are released for production. The following shall be certified by the manufacturer as meeting the requirements of this specification:

Material	(3.1)
Method of manufacture	(3.2)
Color	(3.6.2)
Splices within roll	(3.5.3)
Packaging	(5.1)
Marking of container	(5.2.1)
Length of film on roll	(4.4.1.3.2)

Copies of this certification shall be furnished to the Government Inspector concerned and the procuring agency.

Sample rolls may be tested for more than one requirement. Failure of the test roll to conform with any requirement previously evaluated does not present the testing of that roll for additional defects.

4.3.2 Sampling for production lot acceptance tests. The Government Inspector shall select film rolls for testing in accordance with MIL-STD-105, unless otherwise specified in the appropriate detail description of the tests involved. Unless otherwise approved, no film shall be accepted as a production lot until the pilot lot has been tested and found to conform to this specification.

4.3.3 Inspection for production lots. Inspection for production lots may be made at the manufacturer's plant by the Government Inspector, or at such other place as the procuring agency may specify.

4.4 Tests.

4.4.1 First article lot and periodic production lot tests.

4.4.1.1 Thickness. Unroll a section of film 7.62 meters (25 feet) or more, measure and record the thickness of the unrolled section at twenty points selected at random along the film strip. Determine the average of the twenty measurements. Select five places along the film strip and measure the thickness of the center and both edges at these five points across the face of the film. Grade any roll defective that does not meet the requirements of Drawing 9240825.

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TABLE II First article lot and periodic production sampling plan

Requirement	Requirement paragraph	Test method paragraph	Sample size (no. of rolls)	Rejection level
Film dimensions	3.4	----	----	----
Thickness	3.4	4.4.1.1	10	One or more defective ratings
Width	3.4.3.5.1	4.4.1.2	10	One or more defective ratings
Length on roll	3.4	4.4.1.3.2	10	One or more defective ratings
Core dimensions	3.5.1	----	10	One or more defective ratings
Roll diameter	3.5.1	4.4.1.3.3	10	One or more defective ratings
Roll width	3.5.1	4.4.1.3.5	10	One or more defective ratings
Roll tightness	3.5.2	4.4.1.3.1	10	Two or more defective ratings
Roll splices	3.5.3	4.4.1.3.4	10	One or two, depending on roll length
Roll telescoped or crimped edges	3.5.4	4.4.1.3.6	10	One or more defective ratings
Film visual appearance	3.6	4.4.1.4	----	----
Extrusion defects	3.6.1	4.4.1.4.1	10	Two or more defective ratings
Color	3.6.2	4.4.1.4.2	10	Two or more defective ratings
Clarity	3.6.3	4.4.1.4.3	10	Two or more defective ratings
Film physical properties	3.7	4.4.1.5	3	Two or more defective ratings
Migration characteristics	3.8	4.4.1.6	4	Any deviation from specification
Wrappability	3.9	4.4.1.7	5	One or more defective ratings
Adhesion	3.10	4.4.1.8	5	One or more defective ratings
Modified Taliani stability	3.11	4.4.1.9	3	Any deviation from specification

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4.4.1.2 Width. Unroll a section of film 7.62 meters (25 feet) or more, and measure and record the width of the unrolled section at twenty or more points selected at random along the film strip. Grade any roll defective that does not meet the requirements of Drawing 9240825.

4.4.1.3 Roll dimensions and characteristics

4.4.1.3.1 Tightness of roll. Secure a cardboard core or pipe with the following dimensions: Outside diameter - not less than 0.24 centimeter (3/32 inch) smaller than the minimum outside diameter of the roll core; Wall thickness - 0.48 centimeter (3/16 inch) minimum; Length - 30.5 centimeters (12 inches) minimum. Fasten this core in an upright position. Insert in the core a sponge rubber cushion pad with the following dimensions: Thickness - 2.54 $\pm$ 0.159 centimeters (1 $\pm$ 1/16 inches). Inside diameter - 0.0794 centimeters (1/32 inch) interference to 0.159 centimeters (1/16 inch) clearance with the outside diameter of the cardboard pipe) outside diameter 9.52 $\pm$ 0.159 centimeters (3-3/4 $\pm$ 1/16 inches). Then insert in place over the core above, a paper core the same size as that used for the film roll to be tested, and allow it to rest over the rubber cushion pad at the base of the vertical inner core.

Insert the roll on the upper end of the vertical core and allow the roll to drop a distance of 20.3 centimeters (8 inches) for 38.1-centimeter (15-inch) rolls and 30.5-centimeter (12-inch) vertical core. Any rolls that do not still meet the requirements of Drawing 9240825 shall be graded defective.

4.4.1.3.2 Film length on roll. Check the marking on the containers for the marked length of film on the roll. This statement by the manufacturer may be verified by unrolling the film and measuring the linear feet of film found on the roll. Since it is very difficult to rewind film satisfactorily after unrolling, any film roll that is checked for length will probably have to be scrapped if the film is not used as it is unreeled. The manufacturer should be notified if any roll length is found to deviate by more than 15.2 meters (50 feet) from the length. Any film rolls sampled by the inspector should be marked on the roll core to allow for the amount of film removed. The roll and measured footage readings should be checked against the requirements of Drawing 9240825, and the Government Inspector and procuring agency should be notified if any shipment is found to be incorrectly marked as to film length on the rolls, or if the film rolls are found to have less footage than the limits specified.

4.4.1.3.3 Roll diameter. Measure the outside diameter of the finished rolls and grade any roll defective that exceeds the limits of Drawing 9240825.

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4.4.1.3.4 Splices. Hold the film roll with the flat side facing a light source and visually inspect by transmitted light. Any splice present will usually show up as a break or discontinuity in the film layers on the roll, and by a spot where the light enters between the edges of the splice. Any splices encountered during the use of the inhibiting film shall be reported to the procuring agency and the inspector concerned. Any rolls found to contain splices in excess of the requirements of Drawing 9240825 shall be graded defective.

4.4.1.3.5 Width of roll. Place the film roll between two parallel plates and adjust the plates to fit against both faces of the roll. Measure the distance between the parallel plates, and grade as defective any roll that does not meet the requirements of Drawing 9240825 for width.

4.4.1.3.6 Telescoped or crimped edges. Visually inspect the film roll for the presence of crimped edges or edges that protrude and are liable to become crimped during handling, packaging or shipping operations. Any roll showing crimped or telescoped edges shall be graded defective as required by section 3.6.4.

4.4.1.4 Visual appearance.

4.4.1.4.1 Extrusion defects.

4.4.1.4.1.1 Fisheyes and gels. Unroll a section 7.62 meters (25 feet) or more and cut five sections 91.4 centimeters (3 feet) long at random from the unrolled film. Mark the fisheyes and gels visible in each three-foot section. Measure their thickness versus the thickness of the surrounding film by means of a micrometer or other suitable thickness gauge capable of accurate measurement to 0.0025 millimeter (0.0001 inch). Count and record the number of fisheyes or gels in each strip with a thickness differential of 0.013 millimeter (0.0005 inch) to 0.064 millimeters (0.0025 inch). Count and record the number of fisheyes or gels with a thickness differential of 0.064 millimeter (0.0025 inch) or greater. If any roll contains a particle with a thickness differential of 0.064 millimeter (0.0025 inch) or more, grade the roll defective. If any three-foot section contains fifty (50) or more particles with a thickness differential in the range 0.013 to 0.064 millimeter (0.0005 to 0.0025 inch), grade the roll from which the strip was taken as defective (section 3.7.1.1).

4.4.1.4.1.2 Bubbles. Unroll a section of film 7.62 meters (25 feet) or more, and cut five sections 0.91 meters (3 feet) long at random from the unrolled film. Examine the cut sections visually for the presence of bubbles. If any bubbles or bubble clusters are observed, measure either the diameters or lengths and widths of these bubbles or clusters and grade any roll defective that fails to meet the requirements of section 3.7.1.2.

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4.4.1.4.1.3 Striations. Unroll a section of film 7.62 meters (25 feet) or more and cut five (5) sections 91.4 centimeters (3 feet) long at random from the unrolled film. Examine the cut sections visually for the presence of striations (continuous or semi-continuous ridges or valleys running lengthwise or across the face of the film). If striations are observed, measure the thickness of the film at the post of striation and on the surrounding film. Grade any roll defective that fails to meet the requirements of section 3.7.1.3.

4.4.1.4.1.4 Laminations. Unroll a section of film 7.62 meters (25 feet) or more and cut five (5) sections 91.4 centimeters (3 feet) long at random from the unrolled film. Take each three-foot strip and tear it lengthwise in a series of "Z" or sawtooth rips (using a pinking shears scissors or similar tool) extending to with 0.64 centimeter (1/4 inch) of each edge of the film. Examine the torn edges for their sections or feather edges that peeled away from the body of the film during the tearing operation. Those sections or edges that peel away from the film body are indications of laminations. Grade any roll defective that shows laminations.

4.4.1.4.2 Color. Cut a small section of film about 30 centimeters (1 foot) from the end of the film roll. Place this section against a white background. Place Kodak Color Compensating Filters CC05Y and CC10Y sandwiched together beside the film specimen and visually compare the color of the sample and the combined filters under daylight lighting. A roll showing color darker than the standard shall be graded defective, as specified in 3.6.2.1.

4.4.1.4.3 Clarity. Cut a strip about 91 centimeters (3 feet) from the end of the film roll. Cut this strip into six pieces, stack them, and place the stack in direct contact with a white typewritten sheet of paper. Press the stack of film firmly against the typed sheet and try to read the typing through the six layers of film. If the letters are not distinctly legible when viewed through the plastic, the roll shall be graded defective in accordance with section 3.6.3.

4.4.1.5 Physical properties. Testing to determine physical properties spedified in 3.7 (Table I) should be carried out by using the following procedures:

Tensile strength	ASTM D882
Elongation	ASTM D882
Specific gravity	ASTM D792
Water absorption	ASTM D570
Soluble Matter Loss	ASTM D570
Weight loss in heating	4.4.1.5.2
Deflection	ASTM D648

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4.4.1.5.1 Flow temperature. An injection molding press will be used to produce test pellets for use according to the method described in Procedure A, ASTM D569.

4.4.1.5.2 Weight loss in heating.

(a) Take a 2.54 x 7.62 centimeter (1 x 3 inch) film strip.
(b) Cut 3 test specimens per roll.
(c) Condition for 48 hours @ 23 $\pm$ 1° (73.5 $\pm$ 2°F) and 50 $\pm$ 4% relative humidity.
(d) Weigh to 4 decimal places.
(e) Hang specimens on hooks in circulating oven @ 105 $\pm$ 2°C for 24 hours.
(f) Remove from oven and place in a desiccator with $CaCl_2$ to cool.
(g) Reweigh and express weight loss in mg/cm^2.
(h) Calculations:

$$\text{wt. loss, } mg/cm^2 = \frac{\text{(conditioned wt, g - oven-dried wt, g)}}{\text{area of specimen surface, } cm^2} \times 1000$$

Note: use both top and bottom surfaces for area determination

4.4.1.6 Migration Characteristics.

4.4.1.6.1 Procedure for Standard Plasticizer-Migration Screening Test.

4.4.1.6.1.1 Materials. The inhibitor plastic to be tested should be in the form of a film, not more than 0.25 mm (0.010 inch) thick. The propellant (N-5,) should be prepared as lathe turnings, 0.04 mm (0.0015 inch) thick.

4.4.1.6.1.2 Procedure. A 30 to 40-mg piece of the plastic is weighed, then placed in vacuum desiccator and pumped continuously with a good mechanical pump for one hour. The loss in weight is a measure of the volatile materials in the film. The same piece is then placed in a 20 x 150 mm test tube containing a weight of N-5 propellant turnings equal to twice the weight of the film (Fig. A). The test tube is then necked down (Fig. B) and a piece of 7-mm tubing provided with two constrictions sealed as on Fig. C. This unit is evacuated for one hour, either alone or in manifold with others, by a pump capable of reducing the pressure to one micron or below, then sealed off at the upper constriction (Fig. D). After three days of storage completely immersed in an oil or water bath thermostatically controlled at 40 $\pm$ 1°C, the tube is removed, and the vacuum released at once by breaking the 7-mm tube at the second constriction. The tube is then opened, the propellant and inhibitor weighed, and the inhibitor film analyzed polarographically for nitroglycerin content. Each film sample is run in duplicate, and results reported as "Weight Gain" and "Nitroglycerin Gain," both in percent of the original film weight, after all volatile matter has been removed.

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4.4.1.6.2 Nitroglycerin Determination

4.4.1.6.2.1 Inhibitor samples are dissolved in warm 95 percent ethanol. An aliquot of the diluted ethanol solution is polarographed in a 0.025 molar tetramethyl ammonium chloride solution (75 percent ethanol). The concentration of nitroglycerin is read from a standard graph.

4.4.1.6.2.2 Special Apparatus.

Sargent Model XXI Polarograph, or equivalent.
25-ml volumetric flasks.
30-ml beakers.
Hot plate.
Constant-temperature bath.
Dropping-mercury electrode assembly. NOTE 1.
Cylinder of dry nitrogen gas (containing little or no oxygen). NOTE 2.

4.4.1.6.2.3 Reagents.

Pure nitroglycerin (distilled from Kieselguhr dynamite).
Ethyl cellulose, of the same source as the unknown samples.
95% ethanol.
Redistilled mercury.
0.1 molar tetramethyl ammonium chloride $(CH_3)_4NCl$, Practical Grade, containing 10 ml of 0.1% alcoholic solution of methyl red per liter of final solution.

4.4.1.6.2.4 Preparation of Sample.

Weigh the duplicate inhibitor samples and place into clean 25-ml volumetric flasks.

Add 15 ml of 95 percent ethanol, warm slightly and shake until samples are dissolved. Allow solutions to cool and dilute to volume with 95 percent ethanol. Samples should be in solution, but if a slight turbidity of the solution is observed it will not interfere with the analysis.

4.4.1.6.2.5 Standard Graph.

a. Use an ethyl cellulose sample which is the same type as the unknown sample. NOTE 3. This sample contains no nitroglycerin, but shall be the same in every other respect.

NOTE 1. The dropping mercury electrode assemply consists of a platinum contact electrode, a sintered-glass flushing tube, and a dropping mercury electrode (should have a drop rate of about 6 to 8 seconds per drop at -1.00 volts in the 0.025 molar $(CH_3)_4NCl$ solution). These are held in place by a rubber stopper and the assembly is raised and lowered on a metal rod.

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NOTE 2. The nitrogen gas should contain less than 0.1 percent oxygen. Excess oxygen may be removed by passing the nitrogen gas over absorbents such as hot chromous chloride.

NOTE 3. Experience has shown that any effects of ethyl cellulose on the nitroglycerin polarographic wave are compensated for by using the same type and amount of ethyl cellulose in preparing the Standard Graph as one expects in the unknown samples.

b. Weigh a 30 to 40-milligram sample and place in a clean 25-ml volumetric flask. Add 95 percent ethanol, warm slightly and shake until sample is in solution. Allow to cool and dilute to volume with 95 percent ethanol.

c. Weigh 50 milligrams of pure nitroglycerin into a clean 50-ml volumetric flask that has been previously calibrated. Dilute to volume with 95 percent ethanol.

d. Add 10 ml of ethyl cellulose solution (paragraph b) to a clean 30-ml beaker. Add 5 ml of the pure nitroglycerin solution (paragraph c) to the beaker. Add 5 ml of the 0.1 molar tetramethyl ammonium chloride solution (4.4.1.6.2.3) and stir the contents of the beaker well.

e. Decant about one-fourth of the beaker contents (this will prevent overflow of solution when flushing the sample with nitrogen). Add enough redistilled mercury to the beaker to cover the tip of the platinum contact electrode in the dropping-mercury electrode assembly. Place the beaker into the constant-temperature bath and lower the stopper containing the dropping mercury electrode assembly until it fits well into the beaker. Lower the glass tube for introduction of dry nitrogen into the solution and flush nitrogen through the solution for 5 to 10 minutes. Raise the glass tube and allow nitrogen to flow over the solution during the polarographic analysis. Polarograph the sample from 0.0 volts to -1.60 volts at a sensitivity of 0.060 μa/mm (using the Sargent Model XXI Polarograph). NOTE 1.

f. Three additional points for the Standard Graph are obtained in the same manner (paragraph b) by substituting 4-ml and 2-ml aliquots respectively for the 5-ml aliquot of the pure nitroglycerin solution and adjusting the final volume.

g. The data should show a linear relationship when a plot is made of wave height (current) in millimeters versus milligrams (concentration) of nitroglycerin.

4.4.1.6.2.6 Polarographic Analysis of Sample.

Add 10 ml of the prepared sample (4.4.1.6.2.5, paragraph d) to a clean 30-ml beaker.

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Add 5 ml of 95 percent ethanol and 5 ml of 0.1 molar tetramethyl ammonium chloride solution.

Stir the contents of the beaker well and treat as given in 4.4.1.6.2.5, paragraph e.

NOTE 1. The sensitivity used will depend upon the type of polarograph. This should be adjusted to give about 100-millimeter wave heights.

Obtain duplicate polarograms (current-voltage curves) and refer the average height in millimeters to the Standard Graph. Note 1.

4.4.1.6.2.7 Calculation

The average wave height in millimeters for each inhibitor analyzed, referred to the Standard Graph, will determine the concentration of nitroglycerin present in the solution.

$$\% \text{ nitroglycerine} = \frac{\text{conc of nitroglycerine, mg x total soln vol, ml}}{\text{wt. of sample, mg}} \times 100$$

(See 3.9)

4.4.1.7 Wrappability. Select five rolls at random and use these rolls for evaluation as to the wrappability of the plastic film in inhibiting propellant grains (Ballistite type). Wrap a minimum of 25 grains with the sample film and 30 grains with a previously approved standard film. Wrap the grains in accordance with standard practices under similar operating conditions. Wrapping operations should be observed and inhibited grains should be inspected for quality to insure that they meet the requirements of the applicable drawing and Ordance Classification of Defects. If the sample film is found to be significantly more difficult to process satisfactorily than the standard film, the pilot lot shall be graded unsatisfactory for wrappability. Since this is a subjective test, depending upon the judgement of the operators and observers, the inspection results on the inhibited grains shall be used to assist in judging the performance of the sample film for conformance to the requirements of section 3.10.

4.4.1.8 Adhesion. Select fifteen propellant grains inhibited with the sample films and allow them to cure in open storage at 21-27°C (70-80°F) for two weeks. After curing, test the grains for bond strength in accordance with the procedure given in MIL-STD-286.

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4.4.1.9 Modified Taliani stability. Testing shall be carried out in a closed atmosphere at 93.5°C (200°F) for 23 hours simultaneously by the same procedure, in duplicate, with carefully blended samples (each consisting of 0.25 grams of finely divided chopped ethyl cellulose film and 0.25 grams of the same lot of comminuted N-5 propellant). This test shall be performed for evaluation of each new plasticizer and flake supplier.

4.4.2 Production lot tests. Normal production lot tests may be conducted at the manufacturer's plant by, or under the direction of, the Government Inspector. The Government Inspector shall check each production lot for the properties as specified herein. In addition, the inspector shall satisfy himself that the manufacturer is meeting the requirements for Packaging (5.1) and Marking (5.2.1). The procuring agency may, at its discretion, submit film samples from any production lot received for evaluation of any or all of the properties specified in the first article lot and periodic production lot tests (see 4.4.1), and may reject the production lot on the basis of these tests if the sample film fails to meet any of the test requirements.

4.5 Resubmission. A lot that has been rejected under the provisions of this specification may be resubmitted for inspection and testing, provided the contractor (after being informed of the reason(s) for rejection), submits a signed statement that he has corrected the deficiency or deficiencies noted and has screened the rejected lot to remove the defective material.

5. PACKAGING

5.1 Packaging. Unless otherwise specified, the tape rolls shall be packed in cardboard boxes or drums with a minimum bursting strength of 1.72 MPa (250 psi). The rolls shall be inserted on a core having an outside diameter not less than 1.6 millimeters (1/16 inch) under the minimum roll core diameter specified (wall thickness of core 4.8 millimeters (3/16 inch) or more) and protected from contact with each other and the ends of the container by interlayers of 3.2 millimeters (1/8 inch) thick solid fiberboard. The interlayers shall support the rolls and inner core so that the tape rolls do not contact each other or the sides of the container. A clearance of 1.27 centimeters (1/2 inch) shall be maintained between the face of the tape surface on the outside periphery of the rolls and the inside surface at the sides of the box or container. The weight of a single complete package, including the rolls of inhibiting tape, core, interlayers and the lined container, shall not exceed 34 kilograms (75 pounds). Containers should be lined with a 0.10 millimeter (0.004 inch) polyethylene film sealed with tape.

5.2 Marking

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5.2.1 Marking of Containers. Unless otherwise specified, each container shall be marked with the following information.

Name of manufacturer
Drawing or specification number
Nominal thickness, width and length of film rolls
Contract number
Number of rolls in container
Lot number
Date of manufacturer

In addition, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ethyl cellulose film covered by this specification is for use in inhibiting rocket propellant grain (Ballistite type).

6.2 Ordering data.

6.2.1 Procurement requirement. Procurement documents should specify the following:

(a) Title and number of this specification
(b) Quantity desired
(c) Level of packing (see 5.1)
(d) Color of film required (see 3.3)

Custodian:
Army -AR

Preparing activity:
Army - AR

Review activity:
Army - MI

Project No. 9330-A-885

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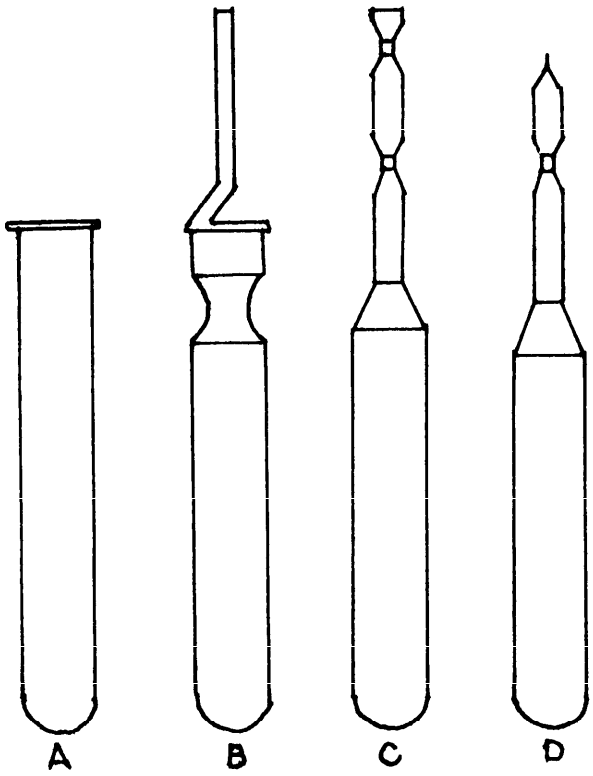


Figure 1. Apparatus for migration tests

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-63463 Film, Inhibiting, Ethyl Cellulose

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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